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Bachelor Thesis Econometrics and Operations Research

Research Proposal

Beyond Tabular Data: Extending Neural Additive Models to Medical Image Recognition via CLIP-Based Concept Features

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1 Research Question

Agarwal et al. (2021) propose NAMs, an inherently interpretable class of neural networks belonging to the family of Generalised Additive Models. A NAM models the expected output as $g(\mathbb{E}[y]) = \beta + \sum_{i=1}^K f_i(x_i)$, where each shape function f_i is a small neural network attending solely to feature x_i . The paper demonstrates near-black-box accuracy on tabular tasks while producing globally interpretable, visualisable shape functions, and explicitly identifies extending NAMs to high-dimensional inputs such as computer vision and language as important future work.

This thesis takes up that call by using CLIP-derived medical concept similarity scores as NAM features, so that each $f_i(x_i)$ describes how the model uses the CLIP-derived similarity score associated with a named clinical concept. The main research question is as follows. Can NAMs using CLIP-derived medical concept similarity scores serve as globally interpretable and accurate medical image classifiers, and do the resulting nonlinear shape functions provide richer model-level concept-response insight than the linear concept-to-class mappings in existing CLIP-based baselines?

Four sub-questions structure the investigation: (i) which post-2021 NAM extensions are most relevant for concept score features and how should they be incorporated; (ii) how does the CLIP-NAM compare in accuracy and interpretability to existing linear concept-bottleneck baselines on the same benchmarks; (iii) what do the shape functions reveal about nonlinear concept effects that linear weights cannot express; and (iv) what are the key limitations regarding identifiability, concavity, and concept selection, and how can they be mitigated. The proposal is explicit that CLIP cosine similarities are model-derived proxies for text-image alignment rather than clinically validated measurements of concept presence; the interpretability claims throughout concern the behaviour of the model with respect to these features, and any clinical interpretation of the learned curves is treated as a separate question requiring expert validation.

2 Relevance, Motivation, and Literature

Relevance

Medical image classification is a high-stakes domain subject to increasing regulatory scrutiny. Under the EU AI Act, AI systems that are themselves medical devices or that serve as safety components of medical devices and fall under medical-device conformity assessment rules can be classified as high-risk, with associated requirements including transparency, risk management, documentation, and human oversight. Dominant post-hoc explanation methods such as Grad-CAM or SHAP are not guaranteed to faithfully reflect a model’s true decision process (Rudin, 2019). The proposed CLIP-NAM pipeline is inherently interpretable by construction: conditional on the input features, the shape functions are an exact description of how the model combines them, rather than a post-hoc approximation. The features themselves are CLIP-derived similarity scores rather than clinically validated measurements, so the construction yields exact model transparency over human-readable proxies of clinical concepts. The research is therefore relevant both to the interpretable machine learning literature and to discussions of model

auditing and regulatory compliance in medical AI.

Motivation and gap

The NAM literature has extended the original model to images, but none of the existing works achieves semantic interpretability of the shape functions. Reuter, Thielmann and Saefken (2024) combine a NAM with a Diffusion Autoencoder to handle image covariates in regression tasks, but the image effect is a single latent sub-network visualised as image morphs rather than named concept scores. Hüls, Perschewski and Stober (2025) apply NAMs to learned convolutional patch features, which carry no semantic label. Wolf, Pölsterl and Wachinger (2023) use an additive architecture combining 3D image and tabular data for Alzheimer’s disease classification with local and global explanations, but the image modality enters as a prototype-based sub-network rather than as a per-concept decomposition over named clinical features.

The concept bottleneck model literature provides semantically grounded image features via CLIP but restricts the prediction head to a linear classifier. Koh et al. (2020) introduce CBMs requiring manual concept annotation. Yang et al. (2023) use a language model to generate candidate concept sentences and align them with images using CLIP, removing the need for supervised concept labels and achieving competitive accuracy across multiple visual datasets; Oikarinen, Das, Nguyen and Weng (2023) similarly eliminate the labelled-concept requirement and scale to ImageNet-like settings. In the medical imaging domain, Patrício, Teixeira and Neves (2024) adapt CLIP to skin lesion classification using concept-based textual descriptions as embeddings, and Yan et al. (2023) apply concept bottleneck models across multiple medical classification tasks including chest X-ray. The prediction head in each of these works is linear and is therefore incapable of capturing nonlinear or threshold effects such as lesion asymmetry being clinically irrelevant below a severity threshold but strongly predictive above it.

The most proximate related work is CAT (Duong et al., 2024), published at ACM SIGKDD 2024, which combines concept representations with a nonlinear Taylor polynomial prediction head. CAT differs from the proposed pipeline in that it operates on grouped tabular features rather than CLIP image-concept similarity scores, uses polynomial rather than NAM shape functions, and has not been applied to medical imaging. It establishes that combining concept representations with nonlinear additive structures is productive; the proposed work specialises this to medical imaging using CLIP-grounded clinical concepts and NAM shape functions. Some concept-based models go beyond linear heads in other ways: Concept Embedding Models (Zarlenga et al., 2022) use higher-dimensional concept representations to mitigate the accuracy-interpretability trade-off, but do not produce global additive shape functions.

To the best of my knowledge, no existing work has combined NAM shape functions with CLIP-derived medical concept similarity scores to obtain global nonlinear concept-level interpretability for medical image classification. The proposed pipeline combines the semantically grounded feature space of CLIP-based CBMs with the globally interpretable additive structure of NAMs.

NAM extensions incorporated

Four post-2021 improvements to the NAM framework are directly relevant here. Siems et al. (2023) propose concurrency regularisation, which penalises pairwise correlations between transformed feature variables in differentiable additive models and is essential because CLIP concept scores for related clinical features are correlated by construction. Xu, Bu, Chaudhari and Barnett (2022) introduce group-sparsity regularisation (SNAM) that shrinks uninformative concept sub-networks, providing automatic concept selection; producing exactly zero sub-networks additionally requires optimisation that supports exact sparsity, for example through proximal updates or thresholding, which is straightforward to add to the training loop. Kim, Choi and Kim (2022) extend NAMs to pairwise interaction terms, allowing the model to capture joint concept effects. X. Zhang, Martinelli and John (2025) provide a systematic analysis of identifiability failures in additive models, showing that multiple shape function configurations can explain the same data differently; their regularisation recommendations directly inform the proposed methodology. Additionally, CAT (Duong et al., 2024) motivates the use of a nonlinear additive head over concept representations; unlike its Taylor polynomial, NAM shape functions impose no restriction on functional form, enabling threshold, saturating, and non-monotone concept effects to be learned and visualised.

3 Methodology and Data

3.1 Replication

The single-task COMPAS classification result from the reference paper is replicated on the COMPAS recidivism dataset (Angwin, Larson, Mattu & Kirchner, 2016), which is the most accessible of the paper’s classification benchmarks given its small size, public availability via ProPublica, and the paper’s inclusion of both a quantitative metric (Table 3 of Agarwal et al. (2021)) and shape function plots that facilitate visual verification of correctness. The replication is implemented in PyTorch. The reference paper evaluates both ExU and standard ReLU hidden units; the appendix of the reference paper lists ReLU as the optimal activation for the COMPAS experiment specifically, and CLIP concept scores are unlikely to produce the highly jagged shape functions for which ExU was designed, so standard ReLU activation is used as the default in both the replication and the extension.

The hyperparameters for the replication are taken directly from Table 4 of the reference paper’s appendix: learning rate 0.02082, output penalty $\lambda_1 = 0.2078$, weight decay $\lambda_2 = 0$, dropout 0.1, feature dropout 0.05, and feature networks consisting of three hidden layers with 64, 64 and 32 units and ReLU activation. The training protocol follows Appendix A.2: 5-fold outer cross-validation, where for each fold the remaining 80 per cent of the data is randomly subsampled 20 times into a 70/10 train/validation split, and the 20 resulting NAMs are ensembled to produce the held-out prediction. Optimisation uses Adam with a batch size of 1024 for a maximum of 1000 epochs with early stopping on the validation fold, and the learning rate is annealed by a factor of 0.995 per epoch. Preprocessing and the treatment of race and gender variables follow the reference paper. Results are reported as AUPRC, consistent with Table 3 of the reference paper, which evaluates binary classification under *area under the precision-recall curve* (denoted

simply “AUC” in the paper because the binary classification datasets are unbalanced); ROC AUC is additionally reported for completeness.

3.2 Datasets

Two publicly available medical imaging benchmarks are used. The **Chest X-ray Pneumonia** dataset (Kermany et al., 2018) comprises 5,856 paediatric chest X-ray images labelled as pneumonia or normal, distributed via Mendeley Data under a CC BY 4.0 licence and mirrored on Kaggle. The class distribution is moderately imbalanced at approximately 73 per cent pneumonia. The **HAM10000** dataset (Tschandl, Rosendahl & Kittler, 2018) comprises 10,015 dermoscopic images across seven diagnostic categories, available through the ISIC archive; the ISIC 2018 Task 3 training data is distributed under a CC BY-NC licence, which permits academic use but not commercial use, and is documented as such in the thesis. More than 50 per cent of HAM10000 lesions are confirmed by histopathology. The class distribution is severely imbalanced, with melanocytic naevi comprising approximately 67 per cent of images; class imbalance is addressed through weighted cross-entropy loss with inverse-frequency class weights. Should the full seven-class task prove infeasible within the thesis scope, a binary reduction to melanoma versus all other categories is applied as a fallback, which is also the most clinically relevant distinction. Both datasets are publicly accessible and are distributed without a formal institutional data use agreement, but their licences and medical-data handling implications are documented explicitly in the thesis.

3.3 Concept set construction and text prompts

The concept set is the set of human-readable text descriptions whose CLIP similarity scores serve as NAM input features. Constructing it is a modelling decision, not a labelling task: no image-level annotation is required. For each dataset, the concept set is built in two complementary steps.

The first step is grounding in established clinical vocabularies. For the chest X-ray task, concepts are drawn from standard radiology reporting terminology: consolidation, pleural effusion, air bronchograms, peribronchial cuffing, bilateral infiltrates, lobar opacity, interstitial pattern, and clear lung fields. For the dermoscopy task, concepts are derived from the ABCD rule (Stolz et al., 1994) and the seven-point checklist (Argenziano et al., 1998): asymmetry, border irregularity, colour variation, differential structures including pigment network and blue-white veil, atypical vascular patterns, and regression structures. This grounding ensures that shape functions are indexed by features whose underlying clinical concepts have external expert validation; the CLIP similarity scores themselves remain model-derived proxies for the visual presence of those concepts rather than direct clinical measurements.

The second step uses a large language model to augment the candidate set. A prompt of the form *“List 20 visual features visible in dermoscopy images that help distinguish melanoma from benign skin lesions”* generates additional candidate concepts beyond the fixed clinical checklists, following the approach of Yang et al. (2023). The LLM name and version, exact prompt wording, query date, number of generated concepts retained, exclusion criteria, and

whether each retained concept has clinical-literature support are documented in the appendix to ensure reproducibility. The augmented candidate set is filtered for clinical plausibility, and the group-sparsity regularisation of the NAM then prunes uninformative concept sub-networks on the training data, yielding a compact predictive subset. Group sparsity selects concepts that are predictive under the model and data distribution; it does not by itself validate clinical relevance.

Each concept is formulated as a natural language sentence following the prompting conventions of Radford et al. (2021): for radiology, “a chest X-ray showing consolidation in the right lower lobe”; for dermoscopy, “a dermoscopic image with an irregular and asymmetric border”. Prompt wording is documented and treated as a design choice subject to sensitivity analysis.

3.4 Feature extraction

For each image, the relevant frozen CLIP encoder computes the cosine similarity score against all K concept text prompts via

$$x_i = \frac{\mathcal{I}(I) \cdot \mathcal{T}(c_i)}{\|\mathcal{I}(I)\| \|\mathcal{T}(c_i)\|} \in [-1, 1], \quad (1)$$

where $\mathcal{I}$ and $\mathcal{T}$ are the image and text encoders respectively. This produces a K -dimensional feature vector per image. For the chest X-ray dataset, BiomedCLIP (S. Zhang et al., 2023) is used as the encoder given its substantially better alignment with biomedical terminology relative to general CLIP, owing to pretraining on approximately 15 million biomedical image-text pairs. For the dermoscopy dataset, standard CLIP (ViT-B/32) is used as the primary encoder; Patrício et al. (2024) report that adapting CLIP using concept-based descriptions improves alignment with ABCDE-rule terminology for skin lesions, so a domain-adapted variant following their embedding strategy is included as a sensitivity check, and a medical/domain VLM baseline is reported alongside the frozen-CLIP results so that the dermoscopy encoder choice is justified empirically rather than by assertion. Both primary encoders are used frozen; no fine-tuning is performed. The resulting feature matrices are stored and used as the sole input to the NAM.

3.5 NAM training and extensions

A NAM is trained on the K -dimensional concept score feature matrices following the architecture of Agarwal et al. (2021). The base NAM already incorporates several regularisation mechanisms, namely mean-centring of each shape function (with a single global bias term carrying the baseline), an output penalty, weight decay, dropout, and feature dropout, together with the ensembling protocol described in Section 3.1. The three extensions of this proposal are layered on top of this stack. First, concavity regularisation (Siems et al., 2023) penalises pairwise correlations between shape function outputs, preventing correlated concepts from redistributing explanatory power arbitrarily. Second, group-sparsity regularisation (Xu et al., 2022) applies a Group LASSO penalty over each sub-network’s parameters; combined with proximal updates this drives uninformative concepts towards exactly zero and produces a compact, automatically selected concept set. Third, pairwise interaction terms (Kim et al., 2022) are added for the top-ranked concept pairs identified after the sparsity step, with the ranking computed exclusively on

the training folds to avoid leaking test information; these terms capture joint effects such as the simultaneous presence of colour variation and border irregularity in melanoma. Because the new penalties are stacked on top of the existing base-NAM regularisation, sensitivity analyses are run over the relative strengths of the existing and added penalties, and the identifiability analysis of X. Zhang et al. (2025) informs both the centring constraints and the choice of regularisation strengths and is used to caveat any interpretation of individual shape functions.

3.6 Evaluation and comparison

Predictive performance is assessed by AUC-ROC, AUPRC, macro-averaged F1 score, sensitivity, and specificity, with balanced accuracy and per-class breakdowns additionally reported for HAM10000. AUPRC, sensitivity, and specificity are particularly relevant given the class imbalance in both datasets and the clinical cost of false negatives in pneumonia detection and melanoma classification. Calibration is reported via reliability diagrams and Expected Calibration Error. The CLIP-NAM is compared against four baselines: LaBo (Yang et al., 2023) (CLIP concepts with a linear classifier), the Label-Free CBM of Oikarinen et al. (2023) (CLIP concepts with a sparse linear classifier), a black-box linear probe on raw CLIP image embeddings, and, for the chest X-ray task, a BiomedCLIP linear probe to ensure the comparison is not biased by encoder mismatch. All baselines are re-implemented under identical data splits, encoders, concept sets, and preprocessing; LaBo and Label-Free CBM were not originally evaluated on these exact medical benchmarks, and clean comparison requires this controlled adaptation rather than reusing results from those papers directly. CAT (Duong et al., 2024) is discussed as a methodological comparator but is not reproduced as an experimental baseline, as it operates on grouped tabular features rather than image-derived concept scores.

An ablation study trains the NAM with each extension (concurvity regularisation, group sparsity, interaction terms) removed in turn to isolate individual contributions. Interpretability evaluation goes beyond visualising shape functions and includes: stability of shape functions across folds and random seeds; sanity checks on the concept-score distributions; correlation and concurvity diagnostics for the final feature set; and a qualitative assessment of whether the learned curves are consistent with patterns documented in the clinical literature, presented as a plausibility review rather than a clinical validation. Examples are reported in which nonlinear shape functions change the predicted class relative to a linear CBM head on the same concept features, isolating the marginal contribution of the nonlinear additive structure. Shape functions are reported as plots of $f_i(x_i)$ against x_i for the top concepts ranked by feature importance, defined as the mean absolute deviation $\frac{1}{N} \sum_n |f_i(x_{ni}) - \bar{f}_i|$ following Agarwal et al. (2021). These plots are the primary interpretability output of the thesis, allowing direct visual assessment of nonlinear concept effects that a linear weight or polynomial coefficient cannot express.

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